FitXchange

Technical Components & Design

Overview

The FitXchange platform is designed to facilitate the exchange of outfits for various social and professional occasions through a robust, database-driven web application. This overview highlights some of the iterative improvements that have been applied to the technical infrastructure. It also demonstrates the application of systematic diagrams into an initial prototype that aims to meet the expectations of our users' goals.

Our development process was driven by user stories, ensuring that the platform aligns with real world needs and interactions. These user stories helped shape the core features and guided the creation of wireframes, which served as blueprints for the interface and interactions.

Technical Infrastructure:

- **Front-End Technologies:** The client-facing side of FitXchange is built using HTML and Pug, a high-performance template engine, which together streamline the creation of HTML markup. JavaScript is extensively used to enhance the interactivity of the web pages, providing a dynamic user experience. This design also employs intermediate CSS styling to improve visibility and device responsiveness.
- **Back-End Technologies:** Node.js serves as the core back-end runtime environment, allowing for scalable server-side scripting some of which specifically target the following areas(dropdown visibility when clicking the search button, listing image formatting and catalogue click behaviours). Docker containers are employed to encapsulate the application's environment, ensuring consistency across development, testing, and production landscapes. We have included more libraries such as npm express-session to facilitate the tracking on user session related behaviours. This implementation has made a significant improvement to support simultaneous users. We developed controllers that act as an interface between our views and models to handle user interactions and business logic, seamlessly integrating the backend with the front-end templates. New libraries involved the use of bcrypt, express NLP Artificial intelligence API connectors.
- **Database Solution:** MySQL, a robust relational database management system, is used to manage structured data efficiently. It supports complex queries and transactions, making it ideal for handling the intricate data relationships within FitXchange. We have made significant updates to the initial design to allow for more streamlined relational database. By stripping away unnecessary tables such as outfits for a more collective structure such as inventory. We

have also reviewed our user stories in more detail and have identified additional tables that will be required by our system.

Application of Systematic Diagrams:

- **Entity-Relationship Diagram (ERD):** Critical for visualizing the structured data in MySQL, the ERD helps define and refine the relational database schema, crucial for effective data management and integrity.
- **Use Case Diagram:** This diagram provides insights into user interactions with the system, clarifying the various functionalities available to different user roles within FitXchange and guiding the user interface design.
- **Sequence Diagrams:** Essential for detailing the interactions between objects over time, sequence diagrams depict the order of operations triggered by transactions, helping developers understand and implement the flow of processes accurately.
- **Activity Diagrams:** These diagrams are used to visualize the flow of operations within specific functionalities, helping to map out the logic behind user activities and system responses, which is vital for coding the back-end services.

Advanced Feature Details

Overview of Planned Advanced Features:

User Login Enhancements: We plan to implement secure authentication by incorporating robust hashing functions for passwords, utilizing bcrypt, and adding cookie management for the “remember me” feature. Cookies will be set in accordance with GDPR guidelines (with appropriate expiry).

Matching Algorithm & Recommendations: An advanced matching algorithm is under consideration to tailor outfit suggestions based on user preferences and historical interactions. Although the algorithm is still in the design phase, its aim is to improve user engagement by recommending outfits that match personal style and event requirements.

User Points/Ratings System: A points system and advanced rating mechanism will be implemented to allow users to rate outfits and experiences. This will incentivize quality listings and create a trust system across the platform.

In-App Messaging: We will develop a basic in-app messaging feature to facilitate communication between users and administrators for dispute management and general queries.

External API Integrations: Planned integrations include using external APIs (such as maps for location-based services and weather data) to enrich the user experience, where relevant.

Implementation Approach:

We will leverage technologies such as Node.js for backend processing, bcrypt for password security, and Chart.js for any administrative statistics. Each feature will be developed iteratively and tested using our CI/CD pipeline.

DevOps and CI/CD Workflows

Docker and CI/CD Implementation:

Docker Build Workflow: A Docker image build workflow has been implemented. This workflow encapsulates the entire development environment, ensuring consistency across development, testing, and production deployments.

GitHub Actions Pipeline: We have set up a GitHub Actions workflow that automates our build process. The pipeline includes:

- Build Stage: Building the Docker image.
- Test Stage: Running automated tests to verify new changes.
- Deployment Stage: Deploying to a staging environment.

These steps contribute to faster iterations and improved deployment consistency.

Personas

Gerald Thompson

Name: Gerald Thompson

"When I set a goal, I give 100% to achieve it. Not immediately, but definitely".



Demographics:

Age: 21-23

Gender: Male

Occupation: Final-year university student

Location: London, UK

Relationship Status: Single

Title: Mr

Education: Bachelor's degree in progress

Goals and Challenges:

- Secure his first professional job
- Feel confident and well prepared for interviews
- Make a lasting positive impression on employers

Motivation

- Wants to make a great first impression in job interviews
- Aspires to kickstart his career with confidence
- Values professionalism and knows attire plays a role in success
- Seeks an accessible solution without financial strain

Frustrations:

- Cannot afford to buy a formal suit on a student budget
- Limited knowledge of where to borrow or rent a suit
- Worried about finding the right fit and looking presentable
- Concerned about comparing with well-dressed candidates who may have more resources.

User Story:

As a final-year student preparing to enter the job market, I want access to a practical and affordable suit borrowing service, so that I can dress professionally for interviews without the financial burden of buying a suit.

User Scenario:

Gerald, a final year student at Roehampton University, is actively applying for jobs in his field. He knows that making a strong first impression at interview is crucial, but buying a new suit is too expensive.

Living in student accommodation with limited storage, he prefers to borrow a well-maintained suit for a short time rather than owing one. He looks for an easy-to-use website that offers a variety of sizes and styles, with clear size guides to ensure the right fit. Affordable or free options with flexible returns, ideally near campus or with online booking, would be ideal.

This service would help Gerald present himself smartly and professionally without financial strain.

Ruby Bennet

Name: Ruby Bennet

"A great outfit can boost confidence. If I can help someone feel their best, why not?"



Demographics:

Age: 27-32

Gender: Female

Occupation: Software Engineer

Location: London, UK

Relationship Status: Single

Title: Miss

Education: Bachelor's degree in software engineering

Goals and Challenges:

- Share her well-maintained outfits with those in need
- Use a secure and structured platform to list her clothes
- Make the lending process simple and hassle-free
- Ensure her outfits are returned in good condition
- Support a sustainable fashion approach

Motivation:

- Helping others feel confident for important events
- Reducing clothing waste by promoting reuse
- Making good use of outfits that would otherwise sit unused

Frustrations:

- Lack of a secure platform that simplifies lending
- Unclear communication with borrowers
- Risk of outfits not being returned or getting damaged
- Websites that are cluttered or difficult to navigate

Environment:

- Lives in a busy city (London), likely in a modern apartment
- Works in an office or hybrid setting at a tech company

User Story:

Ruby is a software engineer working at a tech company. She owns many well-maintained suits and outfits that she no longer needs but would like to lend to others for their special occasions. Ruby is looking for a website where she can list her outfits, allowing people in need to borrow them easily.

User Scenario:

Ruby is a software engineer working at a tech company and a professional in her field. Over the years, she has accumulated many well-maintained outfits. She wants to lend her clothes to those who need them for special occasions. To do this, Ruby is looking for a reliable website where she can securely share her contact information and provide detailed descriptions of her outfits. She needs a well-structured platform that allows her to add the necessary details so other users can easily find and borrow the perfect outfit without confusion.

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Project Documentation

Application Features

- Borrow and lend outfits: users can browse available outfits, request rentals, and track their order list. Additionally, they can list clothing items and set rental conditions.
- User profiles and customization: personalized profiles allow users to manage their preferences, and rental activities.
- Transaction management: the system keeps track of requests, approvals, payments, and returns, ensuring a smooth and transparent process.
- Sustainable fashion initiative: by encouraging outfit sharing, FitXchange reduces clothing waste and supports eco-friendly consumption.

User Stories:

Borrowing & Returning

1. **Browsing and Filtering: As a user** seeking professional or special occasion attire, I want **to browse and filter a collection of outfits**, so that **I can quickly identify pieces that match my style and my event needs**.
2. **Authenticity and Initial Quality Assurance: As a user,** I want **assurance that each outfit is authentic, well-maintained, and ethically sourced**, so that **I can feel confident in my choice and the integrity of the platform**.
3. **Transaction and Rental Process: As a user,** I want **an effortless rental process, including clear pricing, secure payment, and flexible options**, so that **my experience is seamless and hassle-free.**
4. **Delivery and Returning: As a user,** I want **delivery and return information to be clear and accessible,** so that **I can easily understand my options and plan accordingly**.
5. **Fashion Advice and Trends: As a user** unfamiliar with professional or formal dress codes**,** I want **access to fashion tips and trend insights,** so that **I can dress appropriately and stay stylish.**
6. **Reviews and Ratings: As a user,** I want to **leave and read reviews on outfits and lenders** so that **I can make informed decisions about rentals.**

Lending

7. **Creating Listings: As a user** with well-maintained items that are no longer in regular use, I want to **easily create detailed and visually appealing listings**, so that **I can share my outfits with others while ensuring that they're presented accurately.**

8. **Lending Conditions: As a user**, I want to **define lending conditions such as rental duration, pricing, deposit and care instructions,** so that **I can manage risk, protect my belongings and ensure proper use**.

Administrating

9. **User Verification and Security: As an administrator**, I want to **verify new users/user identities,** so that **the platform remains safe, transparent and authentic for all users.**

10. **Moderating and Policy Enforcement: As an administrator**, I want to **effectively manage conflicts and ensure adherence to platform guidelines such as ethical, environmental and intellectual property standards,** so that **disputes are resolved fairly, and the platforms reputation is maintained.**

Account Management

11. **Sign-up and Login: As a user** of the system, I want to **be able to create and log in to my account securely while also being notified of the terms and conditions of the service**, so that **I can make an account and safely become a functioning member of the platform.**

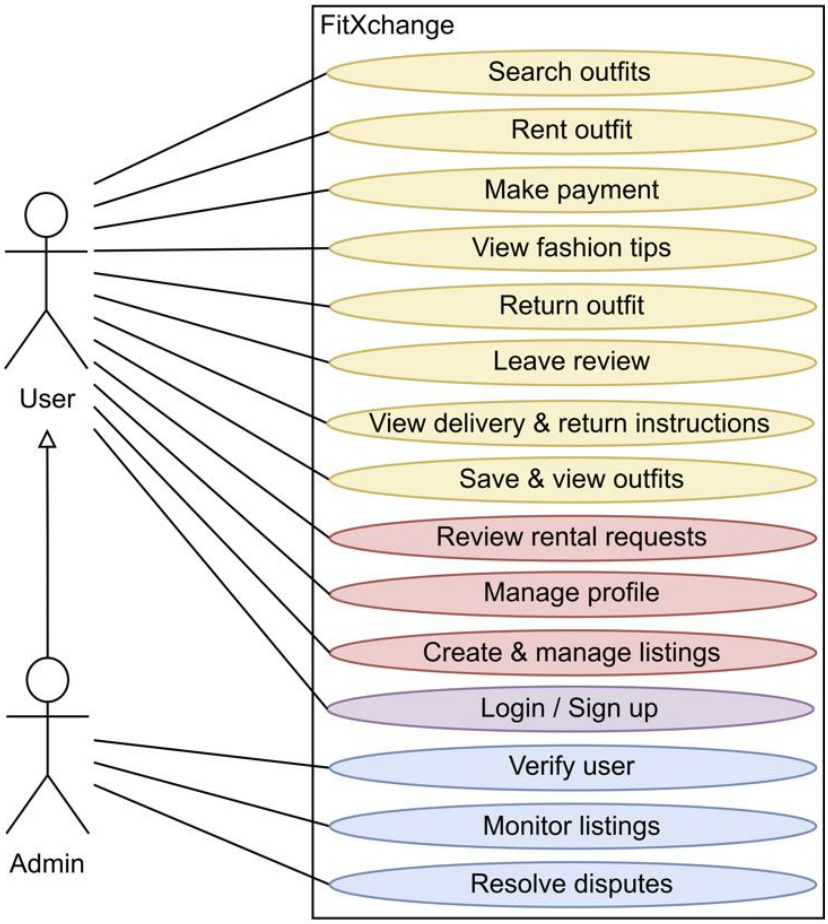
12. **Dashboard and Profile Management: As a user** I want **to edit my profile information, including saved outfits or my listings** so that **I can personalise my experience and keep my dashboard up to date.**

Justification

The user stories were designed to meet the needs of both Gerald Thompson, who values a secure and scalable platform, and Ruby Bennet, who prioritizes an intuitive and user-friendly experience. Borrowers and lenders benefit from seamless browsing, secure transactions, and clear rental conditions, ensuring accessibility while maintaining trust. The platform administrator must be able to verify users and enforce policies with strong security and moderation tools, keeping the platform safe and transparent. Additionally, account management tools ensure seamless onboarding and profile creation, ensuring the platform is efficient, secure, and easy to use.

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Use Case Diagram



Rationale for User Role Generalization:

- **Generalized User Roles:** In FitXchange, while there are theoretically three types of interactions—buying, selling, and administrative—these roles share common attributes and behaviors that allow them to be generalized into two main roles: User and Administrator.
- **User Role:** This role encompasses both buyers and sellers. The decision to generalize these roles into a single 'User' actor in the use case diagram stems from the overlapping functionalities they share, such as browsing items, managing accounts, and processing transactions. A user on FitXchange can seamlessly switch between buying and selling outfits, reflecting the dual nature of their engagement with the platform. This design simplifies the system architecture and reduces complexity by eliminating redundant processes.

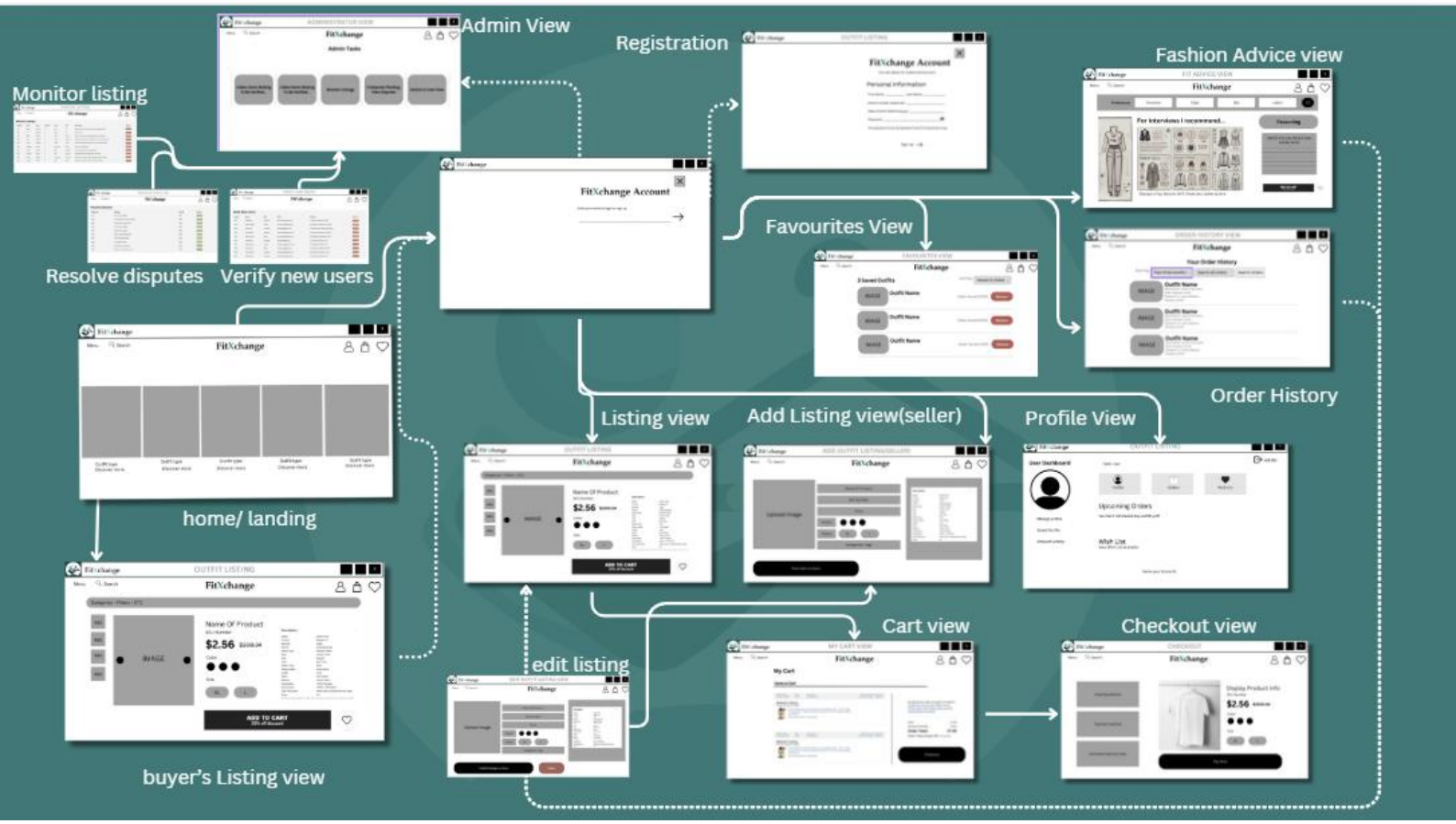
Justification In Further Detail:

- These use cases were refined to ensure that they align closely with the users' needs while also maintaining clarity and efficiency in the platform's interactions. **Browsing and renting** are the central functions, as mentioned in the key features of this project, they allow users to **search outfits**, **view verification** details and **rent** with confidence, eliminating the need for a separate authenticity check. Likewise, the **payment** and **return** processes are streamlined to facilitate smooth transactions, with clear **return and delivery instructions** enhancing user convenience. **Fashion tips** and **reviews** were newly added to enhance user confidence and ensure engagement to the platform by letting users provide guidance and community feedback.
- For **lenders**, the ability to **create and manage listings** ensures that their outfits are being properly represented, while **reviewing rental requests** helps them maintain control over their items.
- Account management was further refined from the last sprint, splitting **profile and dashboard management** while ensuring that **saving and viewing outfits** remain intuitive. The decision to refine memberships and offers under user features helps maintain a just and fair platform.
- On the **administrative** side, the focus remains on **user verification**, **listing moderation** and **resolving disputes,** ensuring security and trust.
- These adjustments ensure that FitXchange remains efficient, user-friendly, secure and meeting both borrower and lender needs without unnecessary complexity.

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Wireframes

For FitXchange, wireframes were instrumental in visualizing the seamless transition between the roles of buyers and sellers, highlighting dual functionality within a single, cohesive user interface. This approach supports the platform's goal of enabling users to effortlessly switch between buying and selling outfits, enhancing user experience and platform efficiency.



Development Integration:

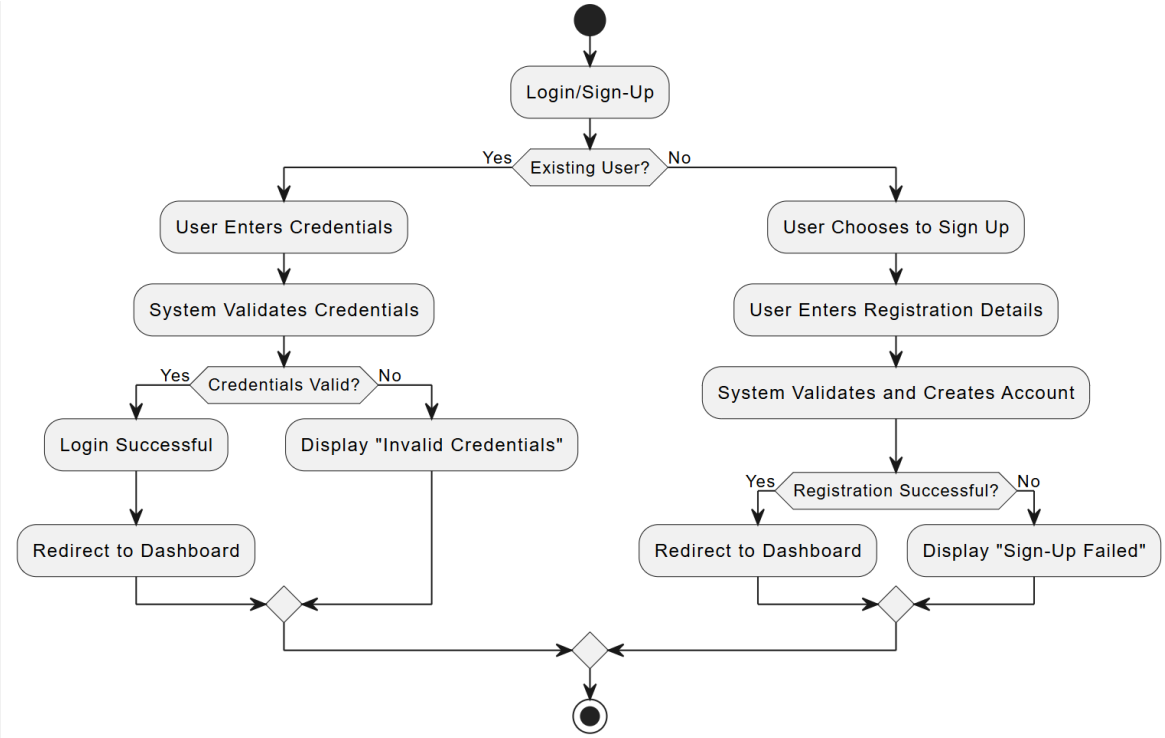
- **Guidance for UI Development:** The wireframes are our a guide to building the front-end code, ensuring that all functional and design requirements are met.
- **Iterative Design and Feedback:** Wireframes will be revised based on feedback from usability testing sessions and stakeholder reviews. This iterative process helps refine the interface for better user engagement and functionality.

Activity Diagrams

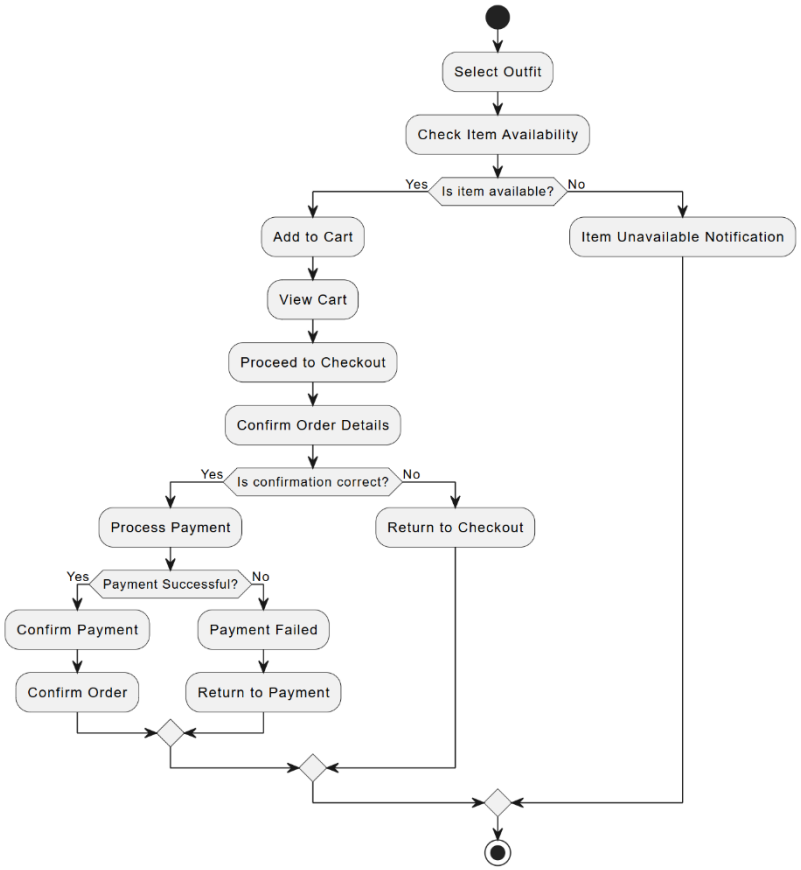
Login/Sign up

Login: User enters credentials, system validates, grant access if valid, or display an error if invalid

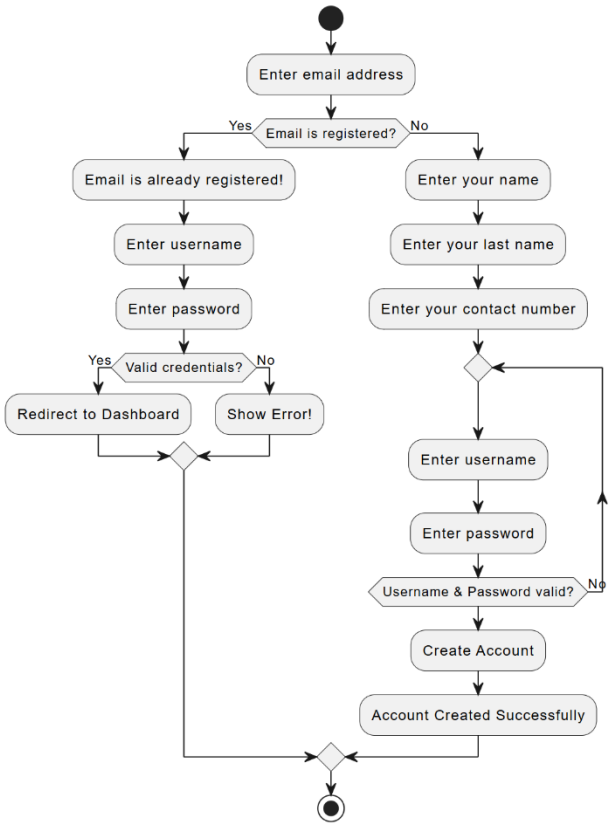
Sign-up: User enters details, system creates accounts, grants access if successful or displays an error if failed.



BORROW/PURCHASE THE OUTFIT	CREATING AN ACCOUNT

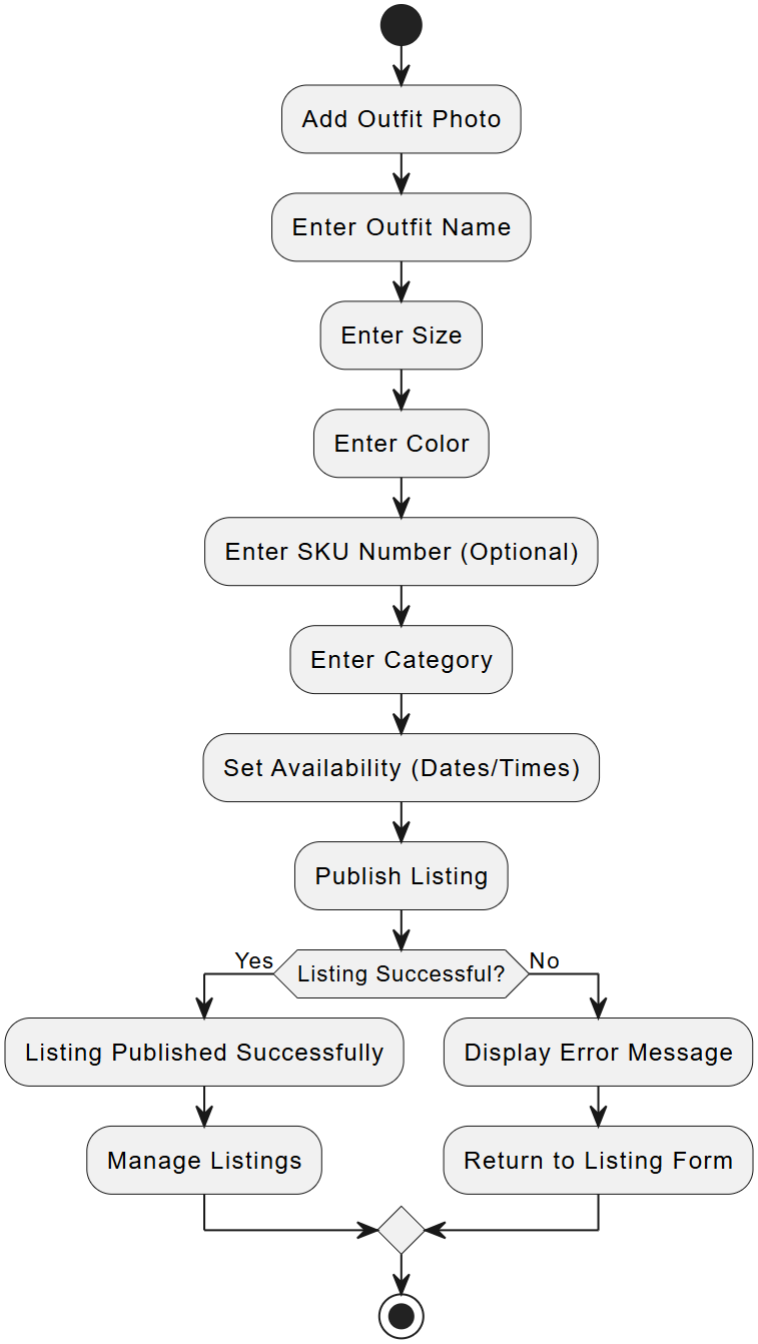


Lending an Outfit: This diagram simplifies the process of listing an outfit for lending. It delineates each step, from user authentication to the final listing approval, pinpointing areas where steps can be consolidated or removed to enhance ease of use. This ensures that users can list outfits quickly and without hassle, increasing the likelihood of frequent user participation and interaction.



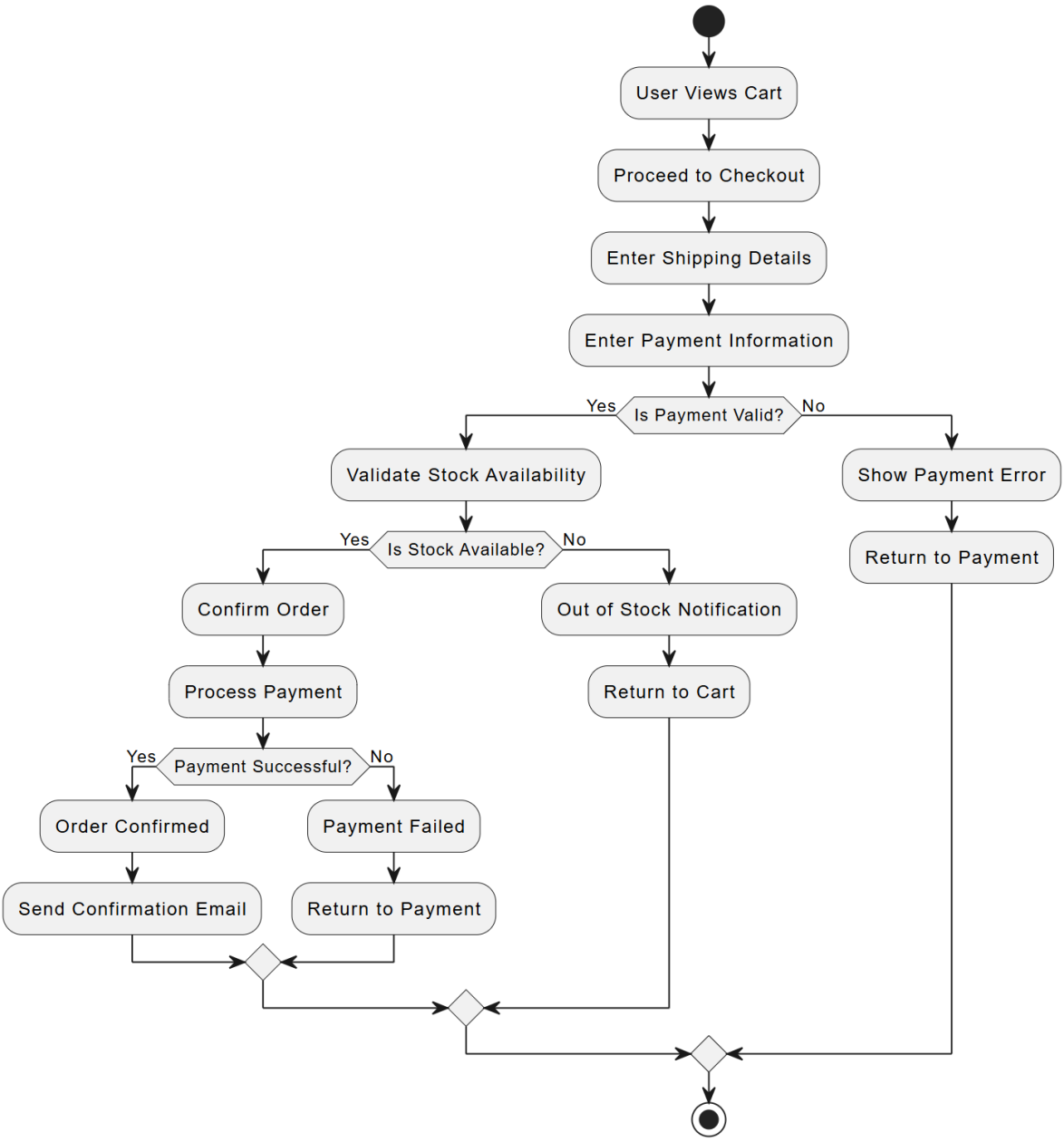
Creating an Account: The activity diagram for creating an account highlights the necessary steps a new user must take to successfully register on the platform. By visualizing this process, we can identify and eliminate any redundant steps and ensure that the account creation is as streamlined as possible, reducing barriers to entry and encouraging more user sign-ups.

LISTING AN OUTFIT



Purchasing an Outfit: To list an outfit on the platform, the lender begins by adding a photo, followed by entering details, setting availability, and publishing. The system then handles success or failure scenarios.

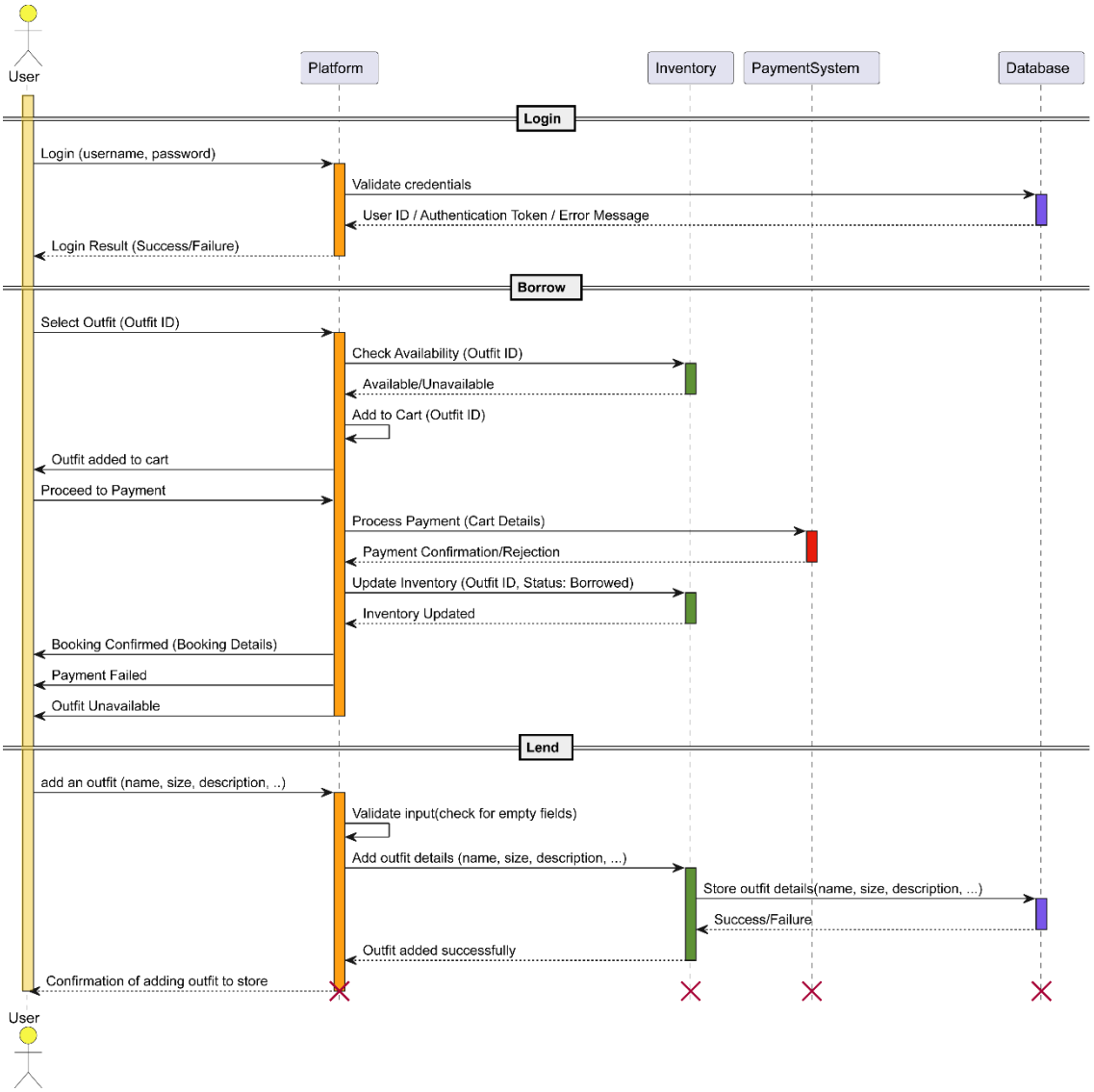
CHECKOUT & PAYMENT



This diagram shows the online checkout process from viewing the cart to order confirmation. It includes steps for entering shipping and payment details, validating payment and stock, processing payment, and handling potential errors like payment failure or stock unavailability.

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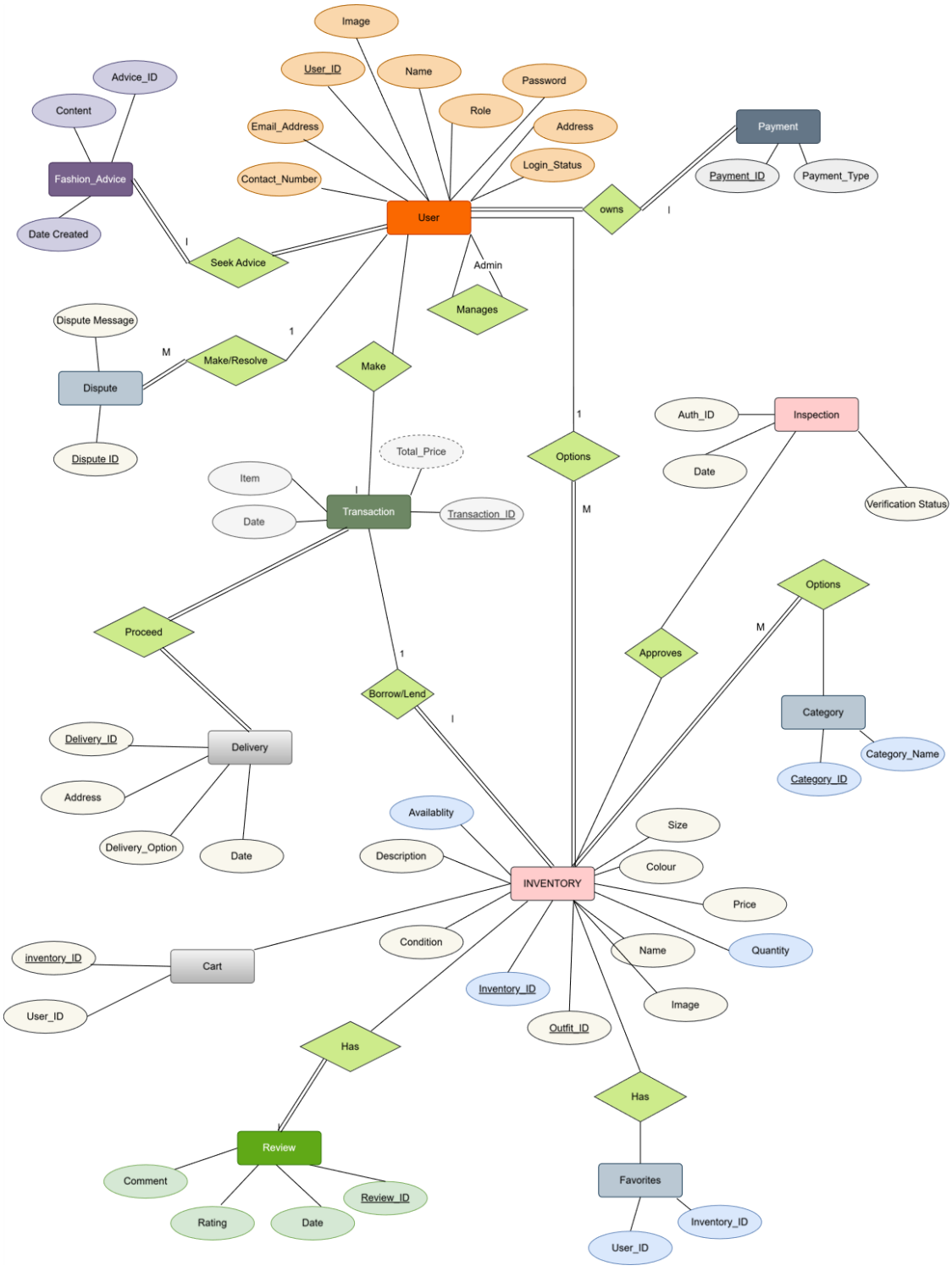
Sequence Diagram



This diagram effectively visualizes how FitXchange processes unfold over time, detailing the communication between objects and the sequence of events that lead to a completed transaction or action. They are instrumental in identifying and approaching dependencies between system components, ensuring that each step in a user interaction is executed efficiently. Given the nature of our service, where users often rent clothes under time constraints, such as for upcoming events or meetings, it is critical that our system processes requests quickly and reliably. Sequence diagrams help us ensure that our platform meets these user expectations by allowing us to streamline operations and minimize response times. This focus on efficiency is not just a technical requirement but a fundamental aspect of user satisfaction, making the sequence diagram a key tool in our development process to enhance overall user experience.

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Entity Relation Diagram



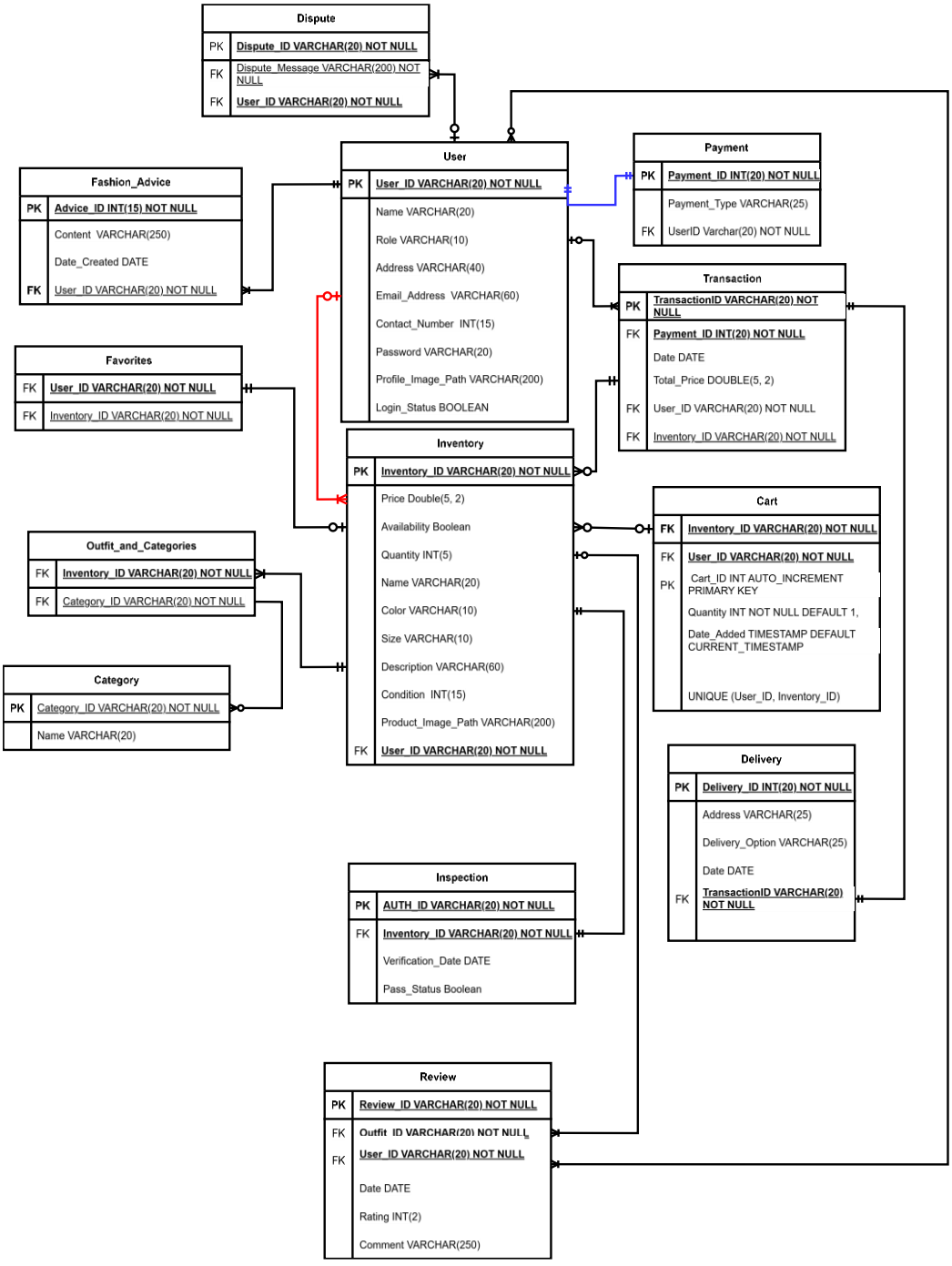
The Entity-Relationship Diagram (ERD) for FitXchange outlines the database schema, ensuring that data is organized and accessed data efficiently for optimal performance.

We anticipate that users may frequently engage in time-sensitive rental transactions for events or meetings. Therefore, it is imperative that our database structure facilitates quick and reliable data retrieval.

The ERD plays a pivotal role in optimizing these processes by clearly defining entity relationships, which helps in streamlining database operations and enhancing query performance. This focus on efficient data management is not merely a technicality but a core aspect of user satisfaction, making the ERD a vital tool in our development arsenal to ensure a superior user experience.

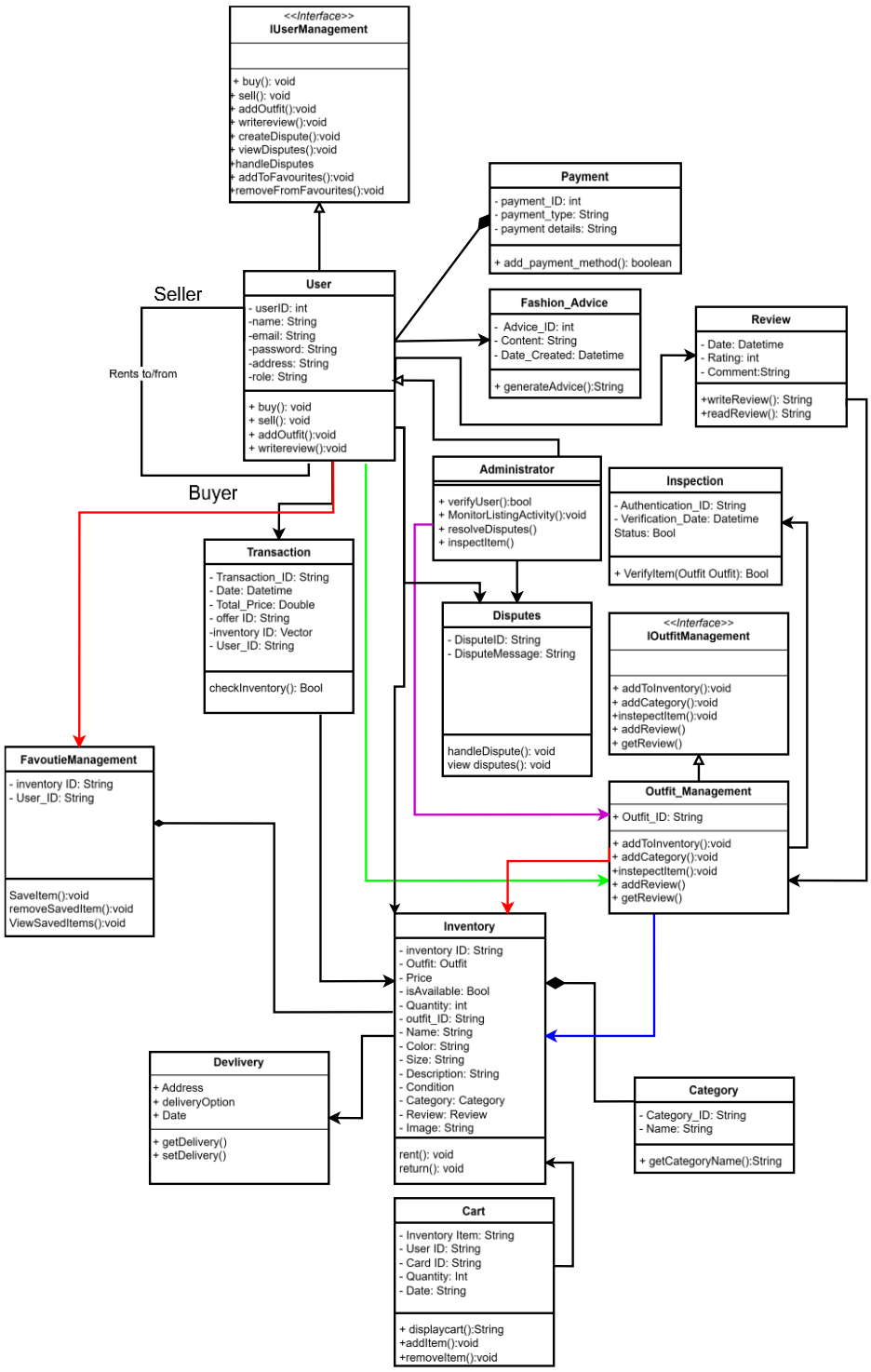
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Physical Data Model



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Class Diagram



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Colour Scheme & Design

